AI BASED SMART SURVEILLANCE SYSTEM

Final Year Project Report



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CERTIFICATIONS

This document has been prepared by all of us together and we take joint ownership of its contents. We have provided references to the material consulted in preparing this document and, to the best of our knowledge, have not plagiarized anything.

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Abstract

Traditional surveillance and access control systems are mostly reactive in nature, relying heavily on continuous human monitoring and conventional authentication methods such as keys, PIN codes, or access cards[1]. These systems often fail to provide intelligent real-time threat analysis and are vulnerable to unauthorized access, theft, duplication, and delayed emergency response. In addition, standard CCTV systems only record events without taking preventive action, which limits their effectiveness in high-security environments. To address these limitations, this project presents an AI-Based Smart Surveillance System capable of performing real-time face recognition, weapon detection, automated access control, and remote monitoring within a unified intelligent framework.

The proposed system is implemented using a Raspberry Pi 4 as the central embedded processing unit. Two artificial intelligence modules operate simultaneously on live video streams. The first module performs face recognition using the Dlib library, which incorporates a CNN-based face detector, facial landmark alignment, and a ResNet-based deep learning model to generate 128-dimensional facial embeddings for identity verification. The second module performs weapon detection using the YOLOv8 convolutional neural network framework to identify suspicious objects such as guns and knives in real time[2]. Both modules are integrated through a multi-threaded Python architecture to ensure parallel execution and efficient decision-making.

The system applies hybrid AND-logic authentication in which door access is granted only when an authorized face is detected and no weapon is identified in the scene. Upon successful authentication, the Raspberry Pi generates a 3.3V GPIO signal that activates a 5V relay module, which in turn controls a 12V electric strike lock for automated door access. Firebase cloud integration enables real-time mobile notifications, remote lock control, event logging, and live video monitoring through a dedicated mobile application.

Experimental testing demonstrates reliable real-time performance on embedded hardware, stable relay and lock operation, accurate face recognition under varying conditions, and effective weapon detection with low latency. The system successfully transforms conventional surveillance from passive monitoring into an active intelligent security solution capable of automated decision-making and threat prevention. The proposed framework provides a scalable, low-cost, and efficient approach for deployment in smart homes, offices, laboratories, educational institutions, and other sensitive facilities, while also offering future expansion toward advanced IoT-based security infrastructures and additional AI-driven threat analysis modules.

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Chapter 1: Introduction

1.1 Background of the Project

Rapid advancements in Artificial Intelligence (AI), embedded systems, and Internet of Things (IoT) technologies have significantly transformed modern security and surveillance infrastructures. Traditional surveillance systems mainly depend on CCTV cameras and manual monitoring, where human operators continuously observe video feeds to identify suspicious activities. Such systems are often inefficient because they are reactive in nature and depend heavily on human attention, which may lead to delayed responses, human errors, and unnoticed threats. Similarly, conventional access control systems based on keys, passwords, PIN codes, or access cards are vulnerable to theft, duplication, and unauthorized usage.

In recent years, intelligent surveillance systems capable of automated decision-making and real-time threat analysis have gained significant importance in homes, offices, laboratories, industries, banks, and other sensitive environments. AI-powered systems can process live video streams, identify individuals, detect suspicious objects, and automatically trigger security actions without requiring continuous human supervision. This shift from passive monitoring toward intelligent proactive security systems has become essential due to increasing security threats and the growing need for smart infrastructure.

The AI-Based Smart Surveillance System developed in this project integrates real-time face recognition, weapon detection, cloud communication, automated door access control, and remote monitoring into a single unified platform. The system uses Raspberry Pi 4 as the main embedded processing unit to execute artificial intelligence algorithms locally on edge hardware. A Dlib-based face recognition pipeline verifies authorized individuals, while a YOLOv8-based object detection model identifies weapons such as guns and knives in real time. The final access decision is implemented using logical conditions that ensure door access is granted only when an authorized person is detected and no weapon is present.

The hardware subsystem includes a relay-controlled 12V electric strike lock connected to the Raspberry Pi through GPIO pins. Firebase cloud integration enables remote monitoring, mobile notifications, event logging, and remote lock control through a mobile application. By combining AI, embedded systems, IoT communication, and automated hardware control, the project provides a low-cost, scalable, and intelligent security solution suitable for modern smart environments.

Chapter 1: Introduction

1.2 Motivation

The primary motivation behind this project is the increasing demand for intelligent security systems capable of preventing unauthorized access and responding to threats automatically in real time. Conventional CCTV systems only record video footage and rely on security personnel to manually monitor screens continuously. In many situations, suspicious activities or armed intrusions remain unnoticed until significant damage has already occurred. Furthermore, manual security systems are often inefficient in large-scale environments where constant monitoring becomes difficult.

Another major motivation is the growing concern regarding accessibility and automation in modern security systems. Traditional locking mechanisms can create difficulties for elderly individuals, disabled persons, and users who require fast and convenient access control systems. Smart authentication mechanisms based on biometric verification provide safer and more efficient alternatives compared to physical keys or passwords.

The integration of artificial intelligence into embedded platforms has also created opportunities for developing cost-effective edge-based surveillance systems that can perform real-time processing without depending entirely on cloud servers. Running AI models locally on Raspberry Pi reduces latency, enhances privacy, minimizes bandwidth usage, and enables faster decision-making for security-critical operations.

In addition, the project aims to bridge the gap between surveillance and automated physical action. Most existing systems are limited to threat detection only, whereas this project combines detection with intelligent access control through relay-operated electric strike locks. The integration of cloud services and mobile applications further enhances usability by allowing users to monitor live video streams, receive alerts, and remotely manage the system from anywhere.

Overall, the motivation behind this project is to develop a smart, intelligent, and affordable surveillance solution capable of improving safety, automation, accessibility, and real-time security response in modern environments.

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1.3 Problem Overview

Traditional surveillance and security systems suffer from several limitations that reduce their effectiveness in preventing unauthorized access and security threats. Conventional CCTV systems are primarily passive systems that only record footage and require human operators to continuously monitor video feeds. This dependency on human observation introduces delays, fatigue-related errors, and reduced efficiency during continuous monitoring operations.

Similarly, traditional authentication systems based on keys, cards, or PIN codes are vulnerable to theft, duplication, unauthorized sharing, and forced access. These methods do not provide intelligent identity verification or threat assessment capabilities. In situations involving armed intrusion or suspicious activity, existing systems often fail to take immediate preventive actions.

Another major problem is the lack of integration between surveillance systems and physical access control mechanisms. Most security systems operate independently without intelligent coordination between monitoring, threat detection, authentication, and hardware-based access management. This separation limits the capability of the system to automatically respond to dangerous situations.

Additionally, many cloud-dependent AI surveillance systems require high computational resources, stable internet connectivity, and expensive hardware infrastructure. These limitations increase system cost and reduce deployment feasibility in small-scale or resource-constrained environments.

To address these problems, this project proposes an integrated AI-Based Smart Surveillance System capable of:

- Real-time face recognition for authorized access verification
- Real-time weapon detection for identifying dangerous objects
- Intelligent decision-making using logical authentication conditions
- Automated electric strike lock control
- Remote monitoring and cloud synchronization through Firebase
- Mobile application-based alerts and access management
- Embedded edge-based AI processing using Raspberry Pi 4

The system transforms traditional surveillance into an intelligent active security framework capable of both detection and automated response.

Chapter 1: Introduction

1.4 Objectives of the Project

The main objective of this project is to design and implement an intelligent AI-powered surveillance system capable of performing automated security monitoring, threat detection, and smart access control in real time.

The specific objectives of the project are as follows:

- To develop a real-time face recognition system capable of identifying authorized individuals using Dlib-based deep learning models.
- To implement a weapon detection system using the YOLOv8 object detection framework for detecting suspicious objects such as guns and knives.
- To integrate both AI modules using a multi-threaded processing architecture capable of simultaneous execution on Raspberry Pi 4.
- To design an intelligent authentication mechanism based on AND-logic where access is granted only if an authorized face is detected and no weapon is identified.
- To implement automated hardware control using Raspberry Pi GPIO pins, relay modules, and a 12V electric strike lock.
- To establish cloud communication using Firebase for real-time alerts, event logging, remote monitoring, and mobile application synchronization.
- To develop a mobile application interface for live video streaming, notifications, and remote lock management.
- To evaluate the performance, reliability, and real-time operation of the integrated surveillance system under different testing conditions.
- To develop a low-cost, scalable, and efficient smart surveillance framework suitable for deployment in homes, offices, institutions, and secure facilities.

1.5 Scope of the Project

The scope of this project includes the design, development, integration, and testing of an AI-driven smart surveillance system capable of performing intelligent access control and real-time threat detection. The project focuses on combining artificial intelligence, embedded systems, IoT communication, and automated hardware control into a single operational framework.

The software scope includes:

- Face recognition using Dlib and ResNet-based facial embeddings
- Weapon detection using YOLOv8 object detection models
- Multi-threaded Python implementation

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- Firebase cloud communication
- Real-time event handling and logging
- Mobile application integration

The hardware scope includes:

- Raspberry Pi 4 embedded processing platform
- USB/IP camera integration
- Relay module interfacing
- GPIO-based hardware control
- 12V electric strike lock implementation
- Power supply integration

The system is designed primarily for indoor surveillance and controlled access environments such as:

- Smart homes
- Offices
- Laboratories
- Educational institutions
- Restricted access facilities

The current implementation mainly focuses on real-time detection, authentication, and automated access control. Features such as large-scale database management, advanced behavioral analytics, facial anti-spoofing, multi-user cloud administration, and large enterprise deployment are outside the current scope and may be considered for future enhancements.

The increasing dependence on digital infrastructure, smart buildings, and automated facilities has created a growing demand for intelligent and reliable security systems. Traditional surveillance solutions mainly rely on CCTV monitoring and manual access control methods such as keys, passwords, or access cards. These systems are limited in their ability to perform intelligent threat analysis and proactive security response. Human operators monitoring surveillance cameras may overlook suspicious activities due to fatigue, distractions, or delayed reaction time. Similarly, conventional authentication systems are vulnerable to theft, duplication, and unauthorized access.

Chapter 2: Problem Definition

The client required a smart security solution capable of automatically verifying authorized individuals, detecting potential threats in real time, and controlling physical access without requiring constant human supervision. The proposed solution needed to be low-cost, reliable, scalable, and suitable for deployment in homes, offices, laboratories, and other restricted-access environments. To fulfill these requirements, the AI-Based Smart Surveillance System was developed using artificial intelligence, embedded systems, IoT communication, and automated hardware integration.

The system combines real-time face recognition, weapon detection, relay-controlled access control, Firebase cloud integration, and mobile application support into a unified intelligent surveillance framework. The implementation follows the requirements initially defined during FYP-I while extending the design and implementation work significantly in FYP-II through hardware integration, AI optimization, cloud communication, and system testing.

2.1 Problem Formulation

Modern surveillance systems face multiple technical and operational limitations that reduce their efficiency in real-world security applications. Traditional CCTV systems only provide video recording functionality and rely heavily on manual monitoring by security personnel. Such systems are passive in nature and cannot independently analyze threats or make decisions in real time. As a result, unauthorized access, suspicious activities, or armed intrusions may remain undetected until serious damage has already occurred.

Similarly, existing access control mechanisms such as keys, access cards, passwords, and PIN-based systems are vulnerable to misuse, theft, duplication, and forced access. These methods do not provide intelligent verification of individuals and cannot assess environmental threats before granting access. In high-security environments, the absence of automated threat analysis significantly increases safety risks.

The client required an intelligent surveillance and access control system capable of:

- Identifying authorized users automatically
- Detecting suspicious objects such as weapons
- Preventing unauthorized access
- Performing real-time decision-making
- Automatically controlling door access
- Providing remote monitoring and alerts
- Operating on affordable embedded hardware

Chapter 2: Problem Definition

To solve this problem, the proposed system integrates two AI modules operating simultaneously:

1. Face Recognition Module
2. Weapon Detection Module

The face recognition module verifies the identity of individuals using Dlib-based deep learning techniques, while the object detection module uses YOLOv8 to detect weapons such as guns and knives in live video streams. The system applies AND-logic authentication where access is granted only if:

- An authorized face is detected
- No weapon is identified in the scene

The final decision is implemented physically through a relay-controlled electric strike lock connected to the Raspberry Pi 4. Firebase cloud integration enables real-time alerts, event logging, and remote monitoring through a mobile application. The system therefore transforms conventional passive surveillance into an intelligent automated security solution capable of both detection and action.

2.2 Mapping to Sustainable Development Goals (SDGs)

The proposed AI-Based Smart Surveillance System contributes directly toward 11th United Nations Sustainable Development Goal (SDG) by improving security, accessibility, technological innovation, and smart infrastructure development.

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SDG 11: Sustainable Cities and Communities

The system contributes toward safer homes, institutions, and workplaces by improving surveillance, reducing unauthorized access, and enabling proactive threat prevention. Intelligent security systems are essential components of modern sustainable smart cities.

By integrating artificial intelligence with automated physical security infrastructure, the project supports the development of safer, smarter, and more accessible environments aligned with modern sustainable development goal.



Chapter 2: Problem Definition

2.3 Record of Meetings with Client

Table 1: Meeting Record

Meeting No.	Date	Discussion Details
1	September 2025	Initial discussion regarding intelligent surveillance system requirements and project scope
2	September 2025	Discussion on real-time face recognition and automated access control
3	October 2025	Finalization of hardware components including Raspberry Pi, relay module, and electric strike lock
4	October 2025	Discussion regarding weapon detection integration using YOLOv8
5	February 2026	Mobile application and Firebase cloud integration requirements
6	March 2026	Hardware integration review and relay-controlled lock mechanism
7	May 2026	System testing, performance evaluation, and final functionality review

Several meetings were conducted with the client throughout the project lifecycle to gather requirements, discuss system functionality, verify implementation progress, and finalize technical specifications. These meetings helped refine the project scope and ensured that the developed system fulfilled the client's security and operational requirements.

Chapter 2: Problem Definition

2.4 Preliminary Product Specifications

The proposed AI-Based Smart Surveillance System consists of both hardware and software components integrated to provide intelligent monitoring, automated access control, and remote security management.

Table 2: Hardware Specifications

Component	Specification
Processing Unit	Raspberry Pi 4 Model B
RAM	8 GB
Operating Voltage	5V DC
GPIO Logic Level	3.3V
Relay Module	5V 4-Channel Relay Module
Locking Mechanism	12V DC Electric Strike Lock
Camera	IP Camera / USB Camera
Communication	Wi-Fi / Internet Connectivity
Power Supply	5V DC for Raspberry Pi, 12V DC for Lock
Storage	MicroSD Card

Chapter 2: Problem Definition

Table 3: Software Specifications

Software Component	Specification
Programming Language	Python
Face Recognition Framework	Dlib + Face Recognition
Object Detection Framework	YOLOv8
Cloud Platform	Firebase
Mobile Application	Flutter Application
Operating System	Raspberry Pi OS
Database	Firebase Fire store
Multithreading	Python Threading

Table 4: AI Model Specifications

Module	Function
CNN Face Detector	Detects human faces in live video
Facial Landmark Predictor	Aligns facial features
ResNet Embedding Model	Generates 128-dimensional facial embeddings
YOLOv8 Detection Model	Detects weapons and suspicious objects

The system is designed to provide real-time processing, low-latency response, reliable hardware control, and scalable deployment capabilities while maintaining low implementation cost.

Chapter 2: Problem Definition

2.5 Expected Functionality of Product

The AI-Based Smart Surveillance System is expected to perform intelligent real-time surveillance, automated threat detection, and secure access control in practical deployment environments. The system continuously captures live video streams through connected cameras and processes them using artificial intelligence models running on Raspberry Pi 4.

The expected operation of the system is as follows:

1. The camera continuously captures live video frames in real time.
2. The face recognition module detects and verifies individuals against the registered authorized database.
3. Simultaneously, the YOLOv8 object detection module analyzes video frames to identify weapons such as guns or knives.
4. The system applies AND-logic authentication:
    - If an authorized face is detected AND no weapon is identified, access is granted.
    - If either condition fails, the system denies access.
5. Upon successful authentication, Raspberry Pi sends a 3.3V GPIO signal to the relay module.
6. The relay module activates the 12V electric strike lock, unlocking the door automatically.
7. Firebase cloud services synchronize system events in real time and send notifications to the mobile application.
8. The mobile application allows users to:
    - View live camera streams
    - Receive real-time alerts
    - Monitor access logs
    - Perform remote lock/unlock operations
9. The system automatically records events and maintains monitoring logs for security tracking and future analysis.

The product is expected to provide a low-cost, intelligent, scalable, and efficient surveillance solution capable of improving safety, automation, and access management in smart environments.

Chapter 3: Problem Analysis

The development of an intelligent surveillance and access control system involves solving a complex engineering problem that requires the integration of multiple engineering domains including embedded systems, artificial intelligence, computer vision, cloud communication, networking, and hardware automation. The proposed AI-Based Smart Surveillance System combines real-time image processing, deep learning algorithms, GPIO-based hardware control, and IoT communication into a unified embedded security framework.

The project requires efficient handling of computational limitations on embedded hardware while maintaining accurate real-time performance. Multiple sub-systems such as face recognition, weapon detection, cloud synchronization, relay control, and mobile communication must operate simultaneously with minimal latency. This chapter analyzes the engineering problem, reviews related projects, highlights the distinguishing features of the proposed system, and evaluates its societal and environmental implications.

3.1 Engineering Problem Model

The proposed project addresses a complex engineering problem involving intelligent security automation and embedded AI-based surveillance. The engineering challenge is not limited to software-based detection only; rather, it involves the integration of artificial intelligence algorithms with physical hardware systems capable of taking automated real-time actions.

The system operates by processing live video streams through two independent AI modules:

1. Face Recognition Module
2. Weapon Detection Module

The face recognition module identifies authorized users using deep learning techniques implemented through the Dlib library. The engineering pipeline involves:

- CNN-based face detection
- Facial landmark alignment
- 128-dimensional facial embedding generation using a ResNet model
- Euclidean distance comparison for identity verification

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The mathematical representation of Euclidean distance used for face matching is:

$$d = \sqrt{\sum_{i=1}^{128}(x_i - y_i)^2}$$

Where:

- (xi) represents the stored facial embedding values
- (yi) represents the detected facial embedding values
- (d) represents similarity distance between two faces

A lower Euclidean distance indicates higher similarity between facial embeddings, enabling accurate identity verification.

Simultaneously, the weapon detection module processes video frames using the YOLOv8 object detection framework. YOLO (You Only Look Once) performs object localization and classification in a single-stage neural network architecture, making it suitable for real-time embedded applications. The model detects suspicious objects such as guns and knives and generates confidence scores for each detected object.

The engineering problem also includes real-time decision-making using logical conditions. The system applies AND-logic authentication:

$$\text{Access} = \text{AuthorizedFace} \land \neg \text{Weapon.Detected}$$

This means:

- Access is granted only if the face is authorized
- AND no weapon is detected

The final decision controls a hardware-based electric strike lock through Raspberry Pi GPIO pins and a relay module. The Raspberry Pi outputs a 3.3V control signal to the relay module, which safely switches a 12V circuit for the electric strike lock using electrical isolation mechanisms.

Chapter 3: Problem Analysis

The project also requires:

- Multi-threaded software execution
- Real-time video processing
- Embedded hardware optimization
- Cloud synchronization through Firebase
- Mobile application communication
- Event logging and monitoring

The engineering complexity arises from integrating all these independent systems into a stable, low-latency, and reliable surveillance framework operating on resource-constrained embedded hardware.

3.2 Recent Similar Projects

Several intelligent surveillance and access control systems have been developed in recent years using artificial intelligence and IoT technologies. These projects mainly focus on face recognition, object detection, or smart monitoring systems individually. However, many existing systems lack complete integration between AI-based surveillance and automated physical access control.

One common category of projects involves AI-based face recognition systems developed using Dlib, OpenCV, and deep learning frameworks. These systems are widely used in attendance management, smart authentication, and identity verification applications. Dlib-based facial recognition systems use CNN face detection and deep feature embeddings to identify individuals with high accuracy. However, most existing systems only perform identity verification and do not include real-time threat analysis or physical hardware automation.

Another category includes object detection systems based on YOLO frameworks such as YOLOv5 and YOLOv8. These systems can detect suspicious objects including guns and knives in surveillance footage. While object detection models provide efficient real-time detection, many implementations operate independently without integrating intelligent access control or user authentication mechanisms.

IoT-based smart lock systems have also been developed using Raspberry Pi, relay modules, and cloud platforms. These systems provide remote monitoring and mobile-based access control but

generally lack advanced AI-based decision-making capabilities. Most existing smart lock solutions rely on passwords, RFID cards, or simple biometric authentication methods.

Cloud-based surveillance systems provide remote monitoring and centralized storage but often depend heavily on internet connectivity and cloud computation. Such systems may introduce increased latency, higher deployment costs, and privacy concerns.

The literature review and analysis of recent projects revealed several limitations:

- Lack of integration between face recognition and weapon detection
- Limited real-time automated physical response
- Absence of intelligent multi-condition authentication
- High dependency on cloud infrastructure
- Limited embedded AI deployment
- Lack of hardware-software synchronization

The proposed AI-Based Smart Surveillance System addresses these limitations by combining:

- Real-time face recognition
- Real-time weapon detection
- Automated relay-based access control
- Firebase cloud communication
- Mobile application support
- Embedded edge-based AI processing
- Multi-threaded decision-making architecture

This integration provides a more intelligent, automated, and proactive security solution compared to many existing systems.

Chapter 3: Problem Analysis

3.3 Distinguishing Features of this Project

The proposed AI-Based Smart Surveillance System contains several distinguishing features that differentiate it from existing surveillance and smart access control systems.

3.3.1 Hybrid AI-Based Authentication

Unlike traditional security systems that rely only on passwords, RFID cards, or face recognition alone, the proposed system combines face recognition and weapon detection simultaneously. The system grants access only when:

- The detected face belongs to an authorized user
- No weapon is detected in the scene

This dual-condition authentication significantly improves security and reduces the possibility of forced or dangerous access scenarios.

3.3.2 Real-Time Embedded AI Processing

Most intelligent surveillance systems rely heavily on cloud computing for AI inference. In contrast, the proposed system performs real-time AI processing directly on Raspberry Pi 4 using embedded edge computing. This reduces:

- Response latency
- Internet dependency
- Cloud processing cost
- Privacy risks

The embedded implementation enables faster local decision-making and improves reliability during network interruptions.

3.3.3 Integration of Detection and Physical Action

Many existing surveillance systems only detect suspicious activities without performing physical actions automatically. The proposed system directly connects AI-generated decisions to hardware automation through:

- GPIO control
- Relay module switching
- Electric strike lock activation

This integration transforms surveillance from passive monitoring into active automated security enforcement.

3.3.4 Multi-threaded Architecture

The system uses multi-threaded Python execution to process:

- Face recognition
- Weapon detection
- Firebase synchronization
- Hardware control
- Mobile communication

simultaneously in real time. This architecture improves system responsiveness and operational efficiency.

3.3.5 Cloud and Mobile Integration

Firebase cloud integration allows:

- Real-time event synchronization
- Live notifications
- Remote lock control
- Access log monitoring
- Mobile application connectivity

Chapter 3: Problem Analysis

The dedicated mobile application enhances user accessibility and enables remote surveillance management.

3.3.6 Low-Cost and Scalable Design

The proposed system is designed using affordable hardware components such as:

- Raspberry Pi 4
- Relay modules
- Electric strike lock
- Standard cameras

This makes the system suitable for practical deployment in homes, offices, laboratories, educational institutions, and small-scale secure facilities.

3.4 Societal and Environmental Implications of the Project

The proposed AI-Based Smart Surveillance System has several positive societal and environmental impacts related to security, automation, accessibility, and technological advancement.

3.4.1 Societal Implications

a) Improved Security and Safety

The system enhances public and private security by providing real-time threat detection and automated access control. The integration of weapon detection with face recognition reduces the possibility of unauthorized or dangerous access into restricted environments.

b) Reduction in Human Monitoring Burden

Traditional surveillance systems require continuous monitoring by security personnel, which may lead to fatigue and operational inefficiency. The proposed intelligent system automates monitoring and decision-making, reducing dependency on constant human supervision.

c) Accessibility Enhancement

Automated contactless access control benefits elderly and differently-abled individuals who may face difficulties using conventional locking systems. Smart authentication improves convenience and usability in modern smart infrastructures.

d) Smart Infrastructure Development

The project contributes toward the development of intelligent smart buildings and automated security systems, supporting technological modernization and safer community environments.

e) Privacy and Ethical Considerations

Although AI-based surveillance improves security, it may also raise privacy concerns related to facial data storage and continuous monitoring. To address this issue, the system stores only authorized user data required for authentication and restricts access to monitoring logs through secure Firebase authentication mechanisms.

3.4.2 Environmental Implications

a) Low Power Consumption

The system uses Raspberry Pi 4, which consumes significantly lower power compared to conventional desktop-based surveillance systems. Edge computing reduces the need for high-power cloud infrastructure and minimizes energy consumption.

b) Reduced Hardware Requirements

The integration of multiple functionalities into a single embedded platform reduces hardware redundancy and minimizes electronic resource usage.

c) Sustainable Smart Automation

The proposed low-cost embedded architecture supports sustainable technological development by enabling efficient and scalable smart surveillance solutions suitable for long-term deployment.

d) Electronic Waste Considerations

As with all electronic systems, improper disposal of hardware components may contribute to electronic waste. Therefore, environmentally responsible disposal and recycling practices should be followed for damaged or obsolete components.

Overall, the project contributes positively toward safer, smarter, and more sustainable environments while promoting intelligent automation and modern security infrastructure.

Chapter 4: Design and Implementation

This chapter describes the complete hardware and software design of the AI-Based Smart Surveillance System along with the implementation techniques used during FYP-II. The system integrates artificial intelligence, embedded processing, IoT communication, and automated hardware control into a unified intelligent surveillance framework. The implementation focuses on achieving reliable real-time performance on Raspberry Pi 4 while maintaining low latency, scalability, and efficient hardware-software synchronization.

The system consists of multiple interconnected modules including:

- Face recognition system
- Weapon detection system
- Relay-controlled electric strike lock
- Firebase cloud communication
- Mobile application interface
- Multi-threaded AI processing architecture

The design combines both software intelligence and physical automation to create a proactive smart security system capable of automated decision-making and access control.

4.1 System Architecture

The overall architecture of the proposed system consists of five major layers:

1. Video Acquisition Layer
2. AI Processing Layer
3. Decision-Making Layer
4. Hardware Control Layer
5. Cloud and Mobile Communication Layer

The cameras continuously capture live video streams and provide input to the Raspberry Pi 4. The AI processing layer executes face recognition and weapon detection simultaneously using deep learning models. The decision-making layer applies AND-logic authentication to determine whether access should be granted or denied. If the detected face belongs to an authorized user and no weapon is detected, the Raspberry Pi activates the relay module through GPIO pins. The relay module then energizes the electric strike lock to unlock the door. Simultaneously, Firebase cloud services synchronize events and send notifications to the mobile application.

Chapter 4: Design and Implementation

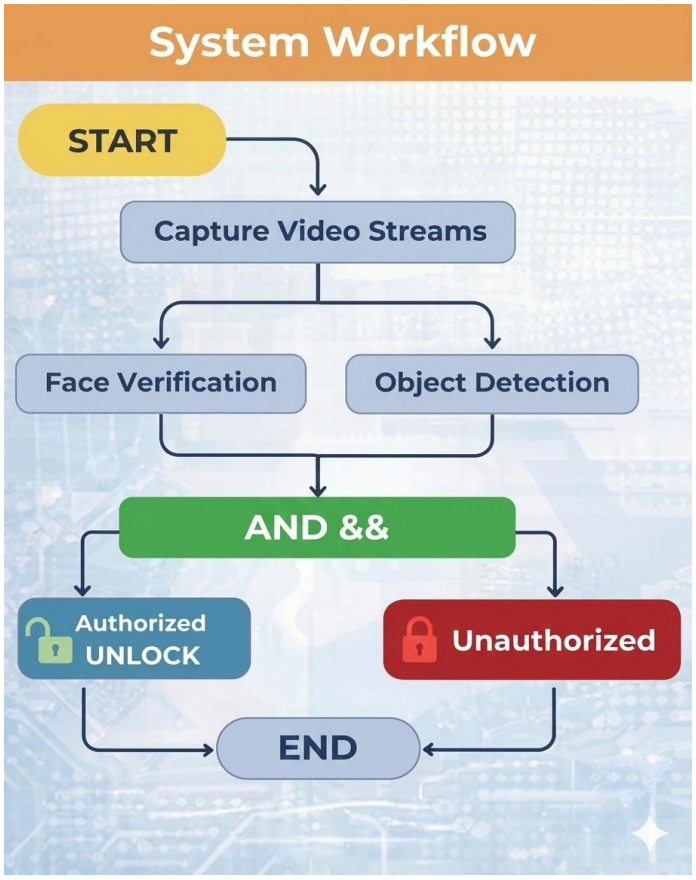


Figure 1: System Architecture

4.2 Hardware Design

The hardware implementation of the system focuses on reliable embedded processing, electrical isolation, automated lock control, and real-time surveillance. The major hardware components include:

- Raspberry Pi 4
- Relay Module
- Electric Strike Lock
- Cameras
- Power Supply Units

Chapter 4: Design and Implementation

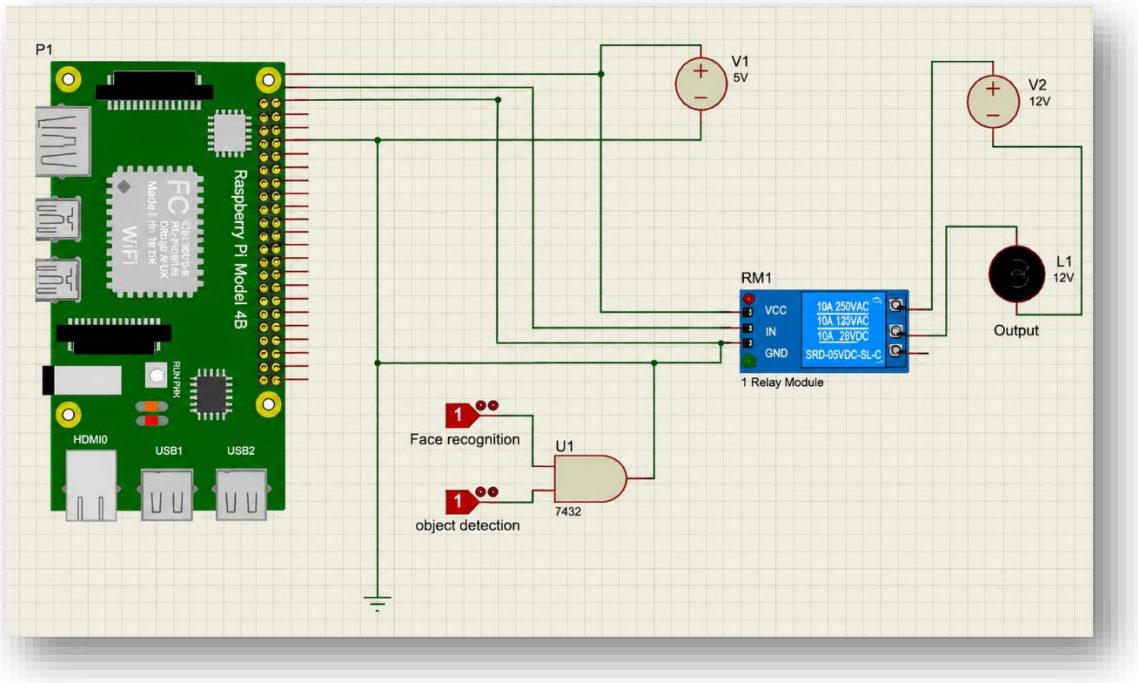


Figure 2: Schematic Design

The Raspberry Pi acts as the central controller that executes AI models and controls all hardware operations. GPIO pins are used for relay activation, while cameras provide live video streams for AI-based processing. The relay module acts as an electrically isolated switching interface between the low-voltage GPIO signals and the high-voltage electric strike lock system. Separate power supplies are used for the Raspberry Pi and locking mechanism to ensure system stability and prevent electrical damage.

Chapter 4: Design and Implementation

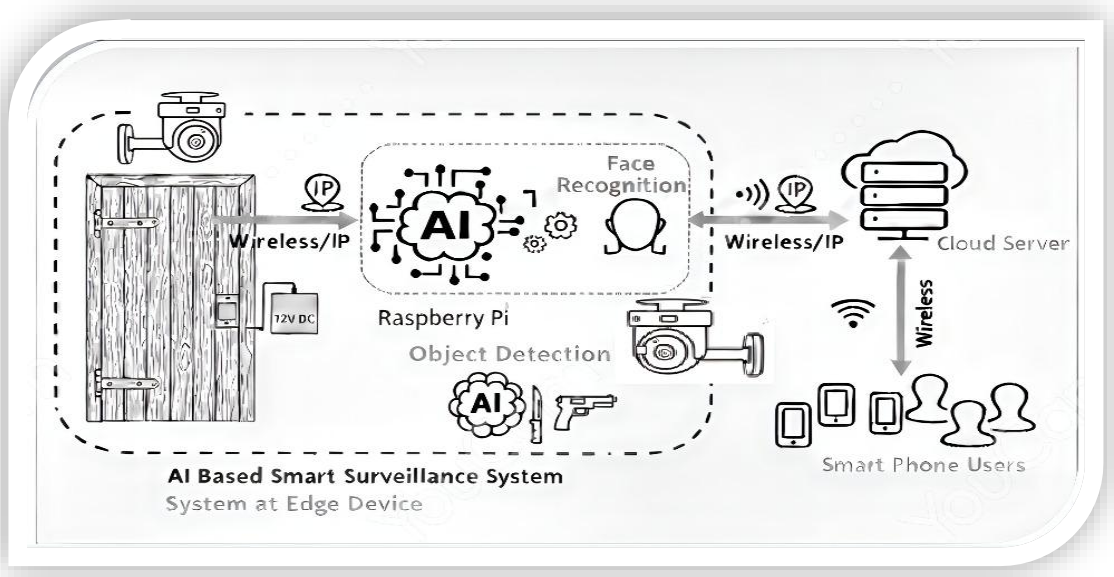


Figure 3: System Design

4.2.1 Raspberry Pi 4 Integration

Raspberry Pi 4 serves as the main embedded processing unit of the system. It performs:

- Real-time face recognition
- Weapon detection and GPIO control
- Firebase communication
- Multi-threaded execution
- Mobile application synchronization

The Raspberry Pi operates on a 5V DC power supply and uses its GPIO pins for hardware interfacing. GPIO17 (Pin 11) is configured as the relay control output pin, while Pin 6 is used as the common ground connection.

The Raspberry Pi internally operates on a 3.3V logic level. When the software generates a HIGH signal on GPIO17, the internal switching circuitry activates and outputs a 3.3V control signal to the relay module.

The system also utilizes:

- USB ports for camera connectivity
- Wi-Fi for Firebase communication
- MicroSD storage for operating system and AI models

The Raspberry Pi was selected due to:

- Low power consumption
- Compact size
- GPIO support
- Linux compatibility
- AI deployment capability
- Cost effectiveness

The embedded edge-processing approach reduces dependency on cloud computation and improves real-time performance.

4.2.2 Relay Module Working

The relay module acts as an electrically isolated switching mechanism between the Raspberry Pi and the electric strike lock. Since the Raspberry Pi GPIO pins operate at 3.3V logic levels and cannot directly drive high-current devices, the relay module provides safe switching of the 12V lock circuit.

The relay module contains:

- Electromagnetic relay coil
- Optocoupler isolation circuit
- Switching transistor
- COM, NO, and NC terminals

The working process is as follows:

1. Raspberry Pi sends a 3.3V HIGH signal to the relay input pin.
2. The optocoupler isolates the Raspberry Pi from the relay power circuit.
3. The transistor activates the electromagnetic coil.
4. The relay armature switches from Normally Closed (NC) to Normally Open (NO).
5. The 12V power supply becomes connected to the electric strike lock.
6. The lock activates and unlocks the door.

The relay module is powered separately using a 5V supply while sharing a common ground with the Raspberry Pi. This design improves electrical safety and protects the GPIO circuitry from voltage spikes.

4.2.3 Electric Strike Lock Mechanism

The electric strike lock is a 12V DC fail-secure solenoid-based locking mechanism used for automated door access control. The lock remains locked during normal operation and unlocks only when electrical power is applied.

The lock contains:

- Solenoid coil
- Metal plunger
- Spring mechanism

The operation is as follows:

1. Under normal conditions, the lock remains mechanically engaged.
2. When the relay completes the 12V circuit, current flows through the solenoid coil.
3. The energized coil generates a magnetic field.
4. The magnetic force pulls the metal plunger.
5. The locking mechanism disengages temporarily.
6. The door unlocks for authorized access.

Once power is removed, the spring mechanism returns the plunger to its original position, automatically locking the door again.

The fail-secure design improves security because the door remains locked during power failure conditions.

4.2.4 Camera Integration

The surveillance system uses cameras for continuous real-time video acquisition. The cameras provide live video streams to the Raspberry Pi for AI-based analysis.

The camera subsystem performs:

- Face image acquisition
- Real-time weapon detection
- Live monitoring
- Video streaming

USB/IP cameras are connected directly to the Raspberry Pi. The cameras continuously capture video frames which are processed by:

- Dlib face recognition pipeline
- YOLOv8 object detection model

The camera configuration was optimized to provide:

- Stable frame rates
- Low latency
- Real-time streaming
- Reliable detection accuracy

Blue USB ports on Raspberry Pi 4 were preferred due to their higher data transfer rates and better support for real-time video streaming.

4.3 Software Design

The software implementation consists of multiple interconnected modules operating simultaneously through a multi-threaded Python architecture. The major software components include:

- Face recognition module
- Weapon detection module
- Firebase communication module
- Mobile application backend
- GPIO control logic
- Event logging system

Chapter 4: Design and Implementation

Python was selected as the primary programming language due to:

- AI library support
- Raspberry Pi compatibility
- Ease of GPIO integration
- Efficient cloud communication support

The software design emphasizes:

- Real-time processing
- Stability
- Modular implementation
- Hardware synchronization
- Low-latency communication

4.3.1 Face Recognition Module

The face recognition system is implemented using:

- Dlib library
- OpenCV
- face_recognition framework

The recognition pipeline consists of three stages:

1. CNN Face Detection

The CNN-based detector identifies human faces in video frames and generates bounding boxes around detected faces.

2. Facial Landmark Alignment

Facial landmark prediction aligns facial features such as eyes and nose to improve recognition accuracy under different orientations.

3. Facial Embedding Generation

A ResNet-based deep learning model generates 128-dimensional facial embeddings representing unique facial characteristics.

The similarity between stored and detected embeddings is computed using Euclidean distance. If the distance remains below a predefined threshold, the face is classified as authorized.

The module provides:

- Real-time verification
- Robustness against lighting variations
- High recognition accuracy
- Partial occlusion handling

4.3.2 Weapon Detection Module

The weapon detection system is implemented using the YOLOv8 object detection framework. YOLOv8 performs:

- Object localization
- Object classification
- Confidence scoring

The model was trained using labeled datasets containing:

- Guns
- Knives
- Other suspicious objects

The system continuously processes live video frames and identifies weapons in real time. If a weapon is detected, the system immediately denies access regardless of facial authentication results.

YOLOv8 was selected because of:

- Fast inference speed
- High detection accuracy
- Real-time processing capability
- Embedded hardware compatibility

The model generates:

- Bounding boxes
- Object labels
- Confidence values

for each detected object.

4.3.3 Firebase Integration

Firebase is used as the cloud communication platform for:

- Real-time event synchronization
- Remote monitoring
- Notification handling
- Access log storage
- Mobile application communication

Firebase Firestore stores:

- Authentication logs
- Detection events
- User activity
- Access timestamps

Firebase Admin SDK was integrated into the Python backend to provide:

- Low-latency communication
- Real-time updates
- Secure cloud synchronization

4.3.4 Mobile Application

The mobile application provides remote access to the surveillance system. The application enables users to:

- View live video streams
- Receive real-time alerts
- Monitor access logs
- Remotely lock/unlock doors

Chapter 4: Design and Implementation

The application communicates with Firebase cloud services to synchronize data with the Raspberry Pi system.

The system also supports:

- Remote lock control
- Push notifications
- Live monitoring updates

through Firebase cloud services.

The mobile application improves:

- User accessibility
- Remote management
- Real-time awareness
- Security monitoring convenience

The frontend interface was designed to provide a simple and user-friendly monitoring environment.

Chapter 4: Design and Implementation

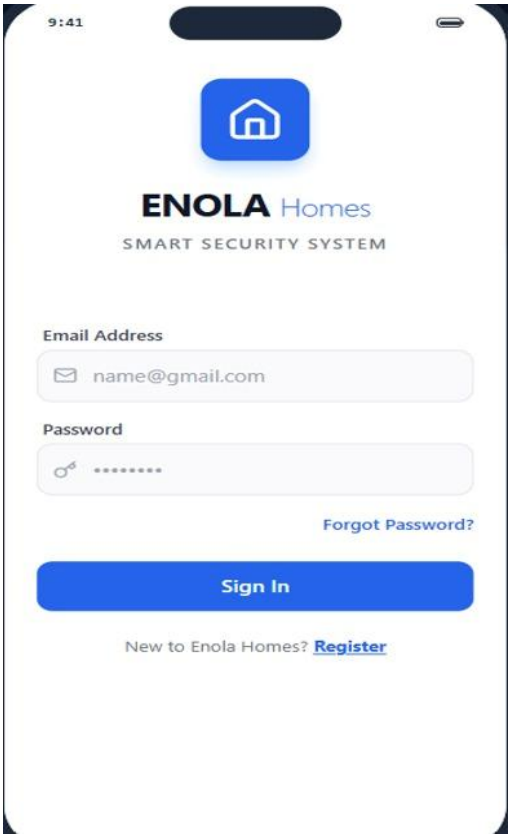


Figure 4: Welcome Screen

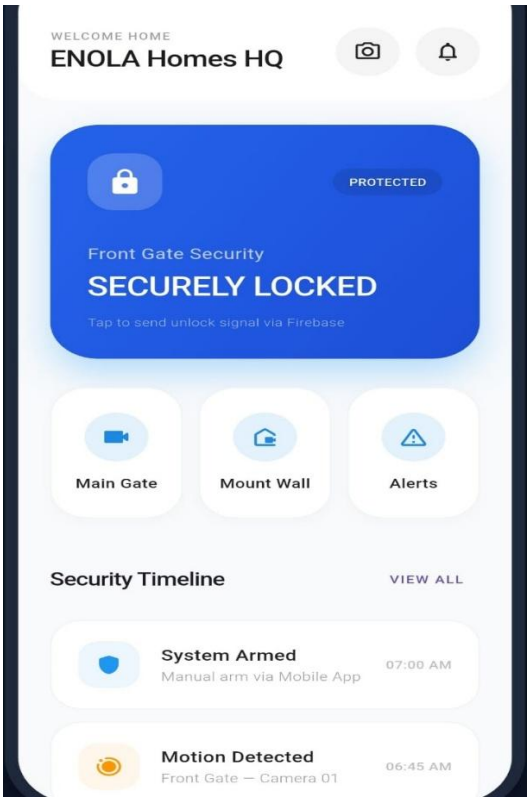


Figure 5: Dashboard Screen

4.3.5 Multi-threaded Processing

The software system uses multi-threaded Python execution to enable simultaneous operation of:

- Face recognition
- Weapon detection
- Firebase communication
- Video streaming
- GPIO control

Multi-threading improves:

- Real-time responsiveness
- Processing efficiency
- Parallel execution
- Reduced system latency

Separate threads continuously monitor:

- Video input streams
- Detection results
- Hardware status
- Cloud synchronization

This architecture ensures smooth operation without blocking critical system functions.

4.4 GPIO and Relay Control Logic

GPIO pins of Raspberry Pi are used for hardware automation and relay activation. GPIO17 is configured as the output control pin connected to the relay module input.

The control logic operates as follows:

- Authorized Face + No Weapon → GPIO HIGH → Relay Activated → Door Unlock
- Unauthorized Face OR Weapon Detected → GPIO LOW → Relay Deactivated → Door Locked

The GPIO output signal operates at 3.3V logic level. The relay module converts this low-voltage control signal into high-voltage switching for the electric strike lock.

The automated control system ensures:

- Fast response
- Reliable hardware switching
- Safe operation
- Secure access management

Chapter 4: Design and Implementation

4.5 Database and Cloud Communication

Firebase Fire store acts as the central cloud database of the system. The database stores:

- Authentication records
- Detection events
- User activity logs
- Access timestamps
- Mobile application updates

Cloud synchronization enables:

- Real-time event updates
- Remote monitoring
- Historical log tracking
- Cross-device communication

The Firebase cloud infrastructure improves:

- Scalability
- Accessibility
- Data synchronization
- System monitoring

All communication between Raspberry Pi and Firebase occurs securely through authenticated API requests.

Chapter 4: Design and Implementation

4.6 Flowcharts and Working Principle

The overall system workflow follows the sequence below:

1. Camera captures live video stream
2. Video frames are processed by AI modules
3. Face recognition verifies user identity
4. YOLOv8 detects suspicious objects
5. Decision engine applies AND-logic authentication
6. GPIO signal controls relay module
7. Relay activates electric strike lock
8. Firebase synchronizes system events
9. Mobile application receives notifications

The system continuously repeats this cycle in real time to provide automated surveillance and intelligent access control.

4.7 Circuit Diagrams and Hardware Schematics

The hardware circuitry includes:

- Raspberry Pi GPIO connections
- Relay module interfacing
- Electric strike lock wiring
- Power supply distribution
- Camera connectivity

Table 5: Circuit connections

Connection	Description
GPIO17 → Relay IN1	Relay control signal
Pin 6 → Relay GND	Common ground
5V Supply → Relay VCC	Relay power
Relay COM/NO → Lock Circuit	Lock switching
12V Supply → Electric Strike Lock	Lock activation

Chapter 5: Investigation and Testing

The investigation and testing phase was conducted to verify that the AI-Based Smart Surveillance System performs according to the required specifications and operates reliably under real-time conditions. Multiple tests were performed on both hardware and software modules to evaluate system stability, detection accuracy, cloud communication, hardware response, and overall operational performance.

The testing process focused on:

- Hardware reliability
- Real-time AI processing
- Authentication accuracy
- Weapon detection performance
- Relay and lock operation
- Cloud synchronization
- Mobile application functionality
- System latency and response time

Each subsystem was tested individually before performing complete integrated system testing.

5.1 Hardware Testing

Hardware testing was performed to verify the proper operation and integration of all physical components including:

- Raspberry Pi 4
- Relay module
- Electric strike lock
- Cameras
- Power supplies
- GPIO connections

The Raspberry Pi was tested for:

- Stable boot operation
- GPIO functionality
- USB camera connectivity
- Wi-Fi communication
- Continuous execution under load

Chapter 5: Investigation and Testing

The relay module was tested to ensure proper switching operation when receiving GPIO signals from the Raspberry Pi. The electric strike lock was tested for stable locking and unlocking response under repeated operations.

Power supplies were monitored to verify:

- Stable voltage levels
- Proper grounding
- Reliable current delivery
- No overheating during continuous operation

The cameras were tested for:

- Continuous live streaming
- Stable frame acquisition
- Real-time video transfer
- Low-latency operation

The integrated hardware setup successfully operated without major electrical faults or communication failures during testing.

5.2 Face Recognition Testing

The face recognition module was tested to evaluate:

- Detection accuracy
- Identity verification performance
- Recognition speed
- Robustness under varying conditions

Testing was conducted using:

- Authorized users
- Unauthorized individuals
- Different lighting conditions
- Multiple face orientations
- Partial facial occlusions

Chapter 5: Investigation and Testing

The Dlib-based recognition pipeline successfully detected and recognized authorized users in real time. Facial embeddings generated through the ResNet model were compared with stored database embeddings using Euclidean distance calculations.

The system demonstrated:

- High recognition accuracy
- Stable real-time performance
- Reliable detection under moderate lighting changes
- Robustness against slight facial orientation variations

The module correctly denied access to unregistered individuals and maintained low false recognition rates during testing.

Sample test observations included:

- Successful recognition within short processing intervals
- Accurate identification for pre-registered users
- Stable operation under continuous video streaming

The face recognition system successfully fulfilled the authentication requirements of the project.



Figure 6: Face recognition

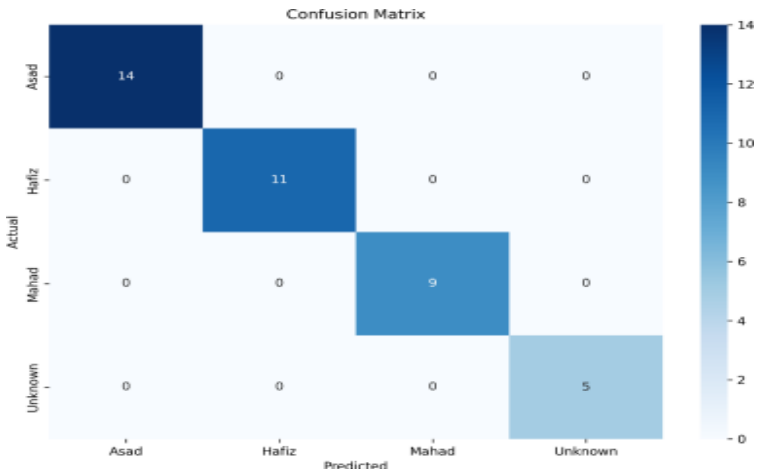


Figure 7: Face recognition confusion matrix

Chapter 5: Investigation and Testing

5.3 Weapon Detection Testing

Weapon detection testing was conducted to evaluate the YOLOv8 model's ability to identify suspicious objects such as:

- Guns
- Knives
- Weapon-like objects

The testing process involved:

- Real-time video input
- Different object positions
- Varying distances
- Multiple lighting conditions
- Simulated threat scenarios

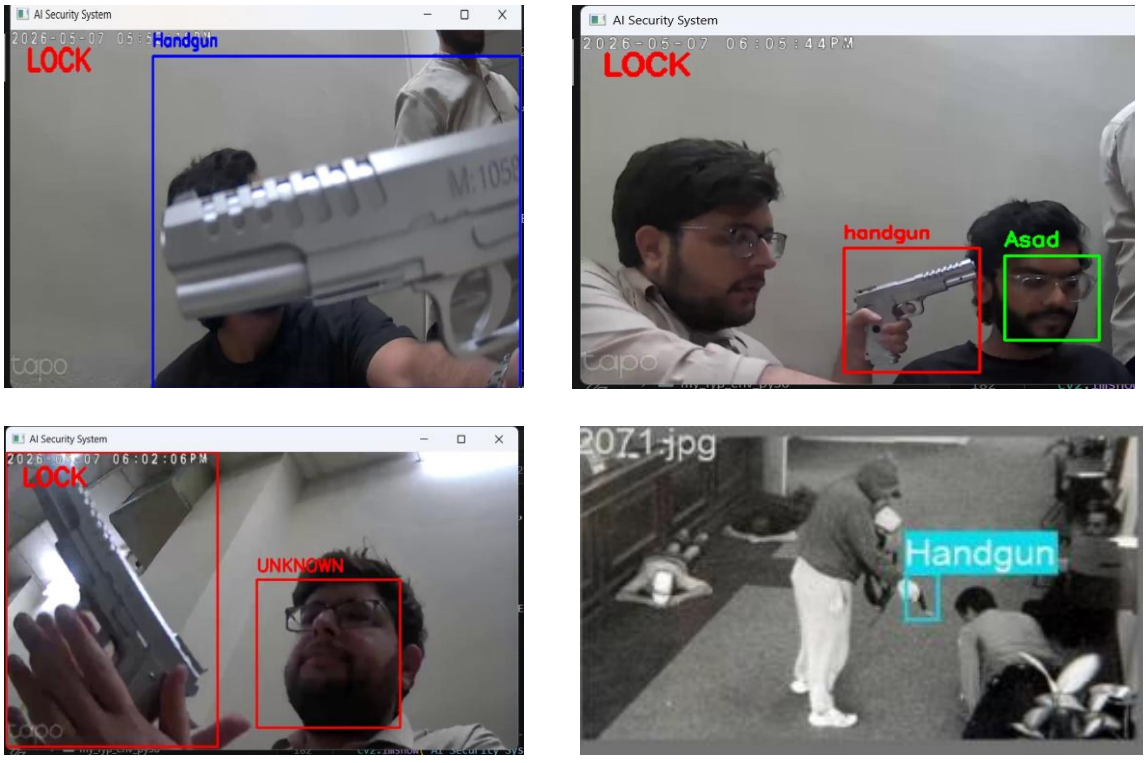


Figure 8: Weapon detection

Chapter 5: Investigation and Testing

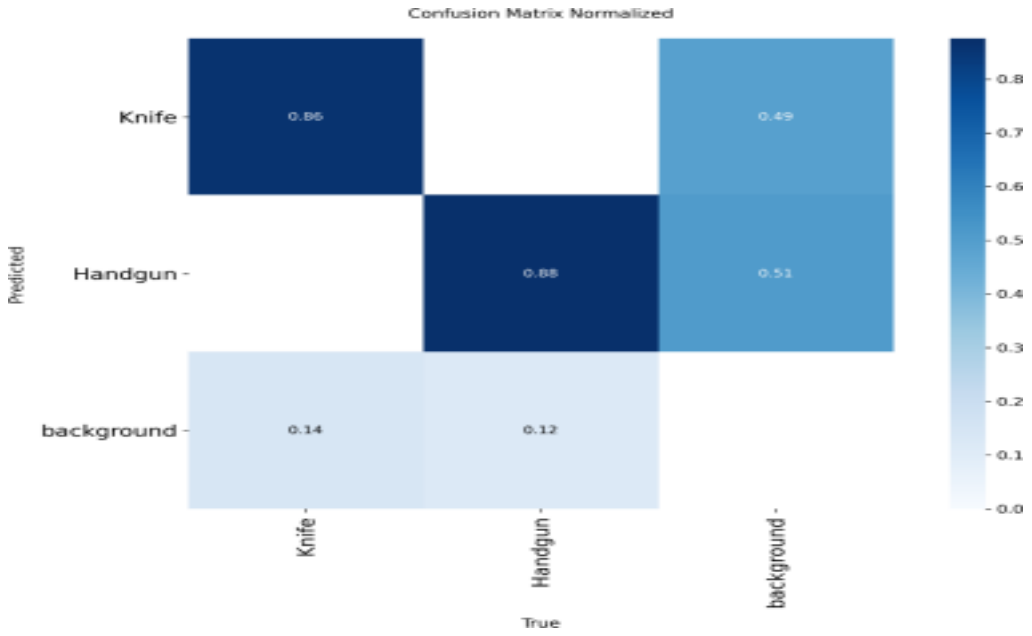


Figure 9: Object detection confusion matrix

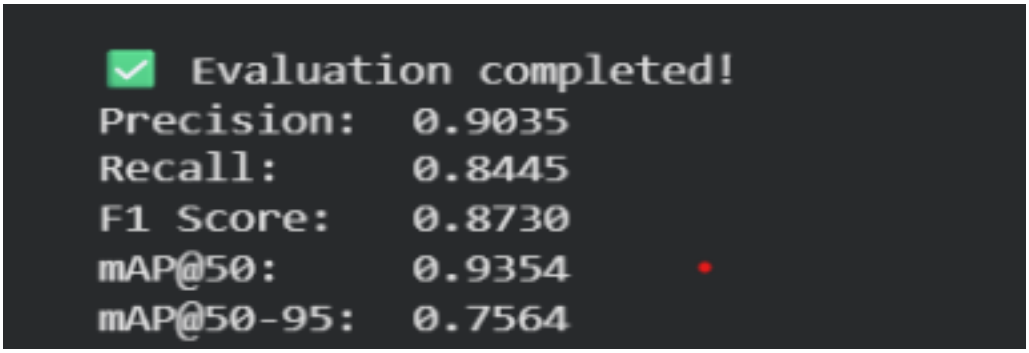


Figure 10: Object detection value scores

Chapter 5: Investigation and Testing

The YOLOv8 model successfully generated:

- Bounding boxes
- Object labels
- Confidence scores

for detected weapons in real time.

The system was tested under:

- Indoor lighting
- Moderate movement conditions
- Single-object detection scenarios
- Multi-object scenes

The weapon detection module successfully prevented access whenever suspicious objects were detected, even if an authorized user was present.

The model demonstrated:

- Fast inference speed
- Reliable real-time operation
- Accurate object classification
- Stable embedded deployment on Raspberry Pi

Minor performance degradation was observed under:

- Extremely poor lighting conditions
- Fast object movement
- Partial object visibility

However, overall system performance remained stable and effective for intended deployment environments.

Chapter 5: Investigation and Testing

5.4 Relay and Lock Testing

Relay and electric strike lock testing were performed to verify:

- GPIO signal transmission
- Relay activation reliability
- Lock switching response
- Automated access control operation

The Raspberry Pi GPIO17 pin was configured to send HIGH and LOW signals based on authentication results.

The testing sequence included:

1. Authorized authentication
2. Relay activation
3. Electric strike lock energization
4. Automatic lock release
5. Door re-locking after signal removal

The relay module successfully switched between Normally Open (NO) and Normally Closed (NC) states when receiving control signals from the Raspberry Pi.

The electric strike lock demonstrated:

- Fast unlocking response
- Stable operation
- Reliable fail-secure locking
- Automatic re-locking after deactivation

Continuous repeated testing confirmed:

- No major switching failures
- Stable relay performance
- Reliable lock activation
- Proper electrical isolation

The integrated hardware control mechanism operated successfully under real-time testing conditions.

Chapter 5: Investigation and Testing

5.5 Firebase Communication Testing

Firebase cloud communication testing was performed to verify:

- Real-time synchronization
- Event logging
- Cloud connectivity
- Remote command transmission
- Mobile notification delivery

The Firebase Admin SDK was integrated with the Raspberry Pi backend to establish cloud communication.

Testing involved:

- Uploading authentication logs
- Sending detection events
- Synchronizing lock status
- Remote access requests
- Mobile alert generation

The system successfully:

- Updated Firestore database entries in real time
- Synchronized access records instantly
- Delivered notifications with minimal delay
- Maintained stable cloud communication

The communication latency remained sufficiently low for real-time monitoring applications.

Internet connectivity interruptions were also simulated during testing to verify system recovery behavior. After reconnection, Firebase synchronization resumed successfully without major data inconsistency issues.

Chapter 5: Investigation and Testing

5.6 Mobile Application Testing

The mobile application was tested to evaluate:

- User interface functionality
- Live video streaming
- Real-time alert reception
- Remote lock control
- Firebase synchronization

Testing scenarios included:

- Authorized access events
- Weapon detection alerts
- Manual lock/unlock commands
- Real-time monitoring sessions

The application successfully:

- Displayed live surveillance feeds
- Received real-time notifications
- Reflected cloud database updates
- Allowed remote lock management

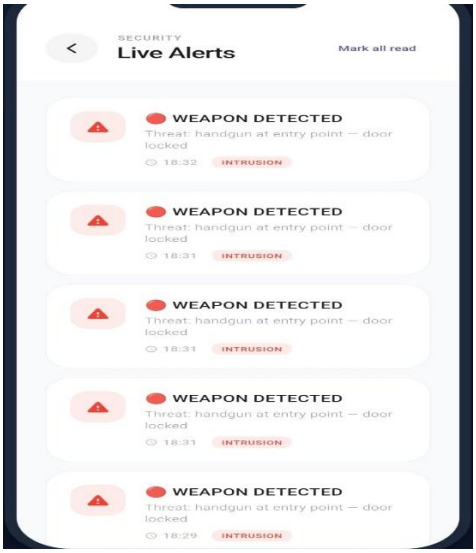


Figure 11: Application Alerts

Chapter 5: Investigation and Testing

5.7 Performance Evaluation

The overall performance of the AI-Based Smart Surveillance System was evaluated based on:

- Detection accuracy
- Processing speed
- Response latency
- Cloud synchronization
- Real-time operational stability

The embedded edge-processing approach using Raspberry Pi successfully achieved:

- Real-time inference
- Continuous video processing
- Stable multi-threaded execution
- Reliable hardware automation

The face recognition module achieved high verification accuracy under normal indoor conditions. The YOLOv8 model demonstrated reliable weapon detection performance with fast processing speeds suitable for real-time surveillance applications.

The integrated system showed:

- Stable continuous operation
- Reliable GPIO response
- Efficient Firebase synchronization
- Effective automated access control

The use of multi-threaded processing improved:

- System responsiveness
- Parallel execution efficiency
- Real-time operational performance

The system successfully satisfied the functional requirements defined during project planning.

Chapter 5: Investigation and Testing

5.8 Experimental Results and Analysis

Experimental analysis was conducted to evaluate the integrated performance of all system modules under practical operating conditions.

Table 6: System Performance

Parameters	Performance
Face Recognition Accuracy	97%
Object Detection Accuracy	93%
Unauthorized Detection Response	1 second
Firebase Communication Delay	< 100ms
Door Unlock Response Time	1-2 seconds

Table 7: System investigation

Test Parameter	Observation
Face Recognition	Accurate identification of authorized users
Weapon Detection	Reliable detection of guns and knives
GPIO Control	Stable relay triggering
Lock Operation	Fast and reliable unlocking response
Firebase Sync	Real-time cloud synchronization
Mobile Alerts	Successful instant notification delivery
Multi-threading	Smooth simultaneous execution
Embedded Processing	Stable Raspberry Pi operation

Chapter 5: Investigation and Testing

The integrated system successfully performed:

- Real-time AI processing
- Intelligent decision-making
- Automated hardware response
- Remote cloud communication

The AND-logic authentication mechanism effectively prevented access whenever:

- Unauthorized individuals were detected
- Weapons were identified

The experimental results demonstrated that the proposed system successfully transformed traditional surveillance into an active intelligent security solution.

5.9 Limitations of the System

Although the proposed system demonstrated reliable functionality, several limitations were identified during testing and deployment.

a. Low-Light Performance

Face recognition and object detection accuracy may decrease under extremely poor lighting conditions due to reduced image clarity.

b. Embedded Hardware Constraints

Running multiple AI models simultaneously on Raspberry Pi introduces computational limitations that may reduce performance under very high workloads.

c. Internet Dependency for Cloud Features

Firebase synchronization and mobile notifications depend on stable internet connectivity. Cloud-related features may become temporarily unavailable during network interruptions.

d. Limited Detection Dataset

The weapon detection model was trained on a limited dataset and may not generalize perfectly to all real-world weapon variations or uncommon object appearances.

e. Single-Environment Optimization

The current implementation was optimized mainly for indoor controlled-access environments and may require additional optimization for large outdoor deployments.

f. Absence of Facial Anti-Spoofing

The current system does not implement advanced anti-spoofing mechanisms to detect fake images or printed photographs.

g. Limited Multi-User Scalability

The system currently supports moderate-scale deployments and may require database optimization for very large user populations.

Despite these limitations, the system successfully achieved its primary objectives and demonstrated the feasibility of implementing a low-cost AI-powered intelligent surveillance and access control solution using embedded edge computing.

Chapter 6: User Guide

This chapter provides a comprehensive guide for operating the AI-Based Smart Surveillance and Security System. The system is designed to enhance security through real-time face recognition, weapon detection, automated door access control, and mobile application integration. By combining artificial intelligence, computer vision, Raspberry Pi edge computing, Firebase cloud services, and a Flutter-based mobile application, the system enables users to monitor security events, receive instant alerts, access live video streams, and manage door operations remotely. The following instructions describe the procedures for system startup, user authentication, live monitoring, access control, alert management, and system shutdown to ensure efficient and reliable operation.

6.1 System Installation and Mounting Guidelines

1. Install the Raspberry Pi 4 inside a secure enclosure near the main entrance of the building.
2. Connect the Raspberry Pi to a stable power supply and network connection.
3. Mount the face recognition camera at eye level (approximately 1.5–1.8 meters above the ground) facing the entrance.
4. Position the weapon detection camera to provide a wider view of the entrance area and surrounding environment.
5. Install the electronic door lock and relay module securely on the door according to the manufacturer's specifications.
6. Connect the relay module to the Raspberry Pi GPIO pins and verify proper wiring connections.
7. Ensure all cables are properly organized and protected to prevent accidental damage or tampering.
8. Install the ENOLA Security mobile application on authorized Android devices.
9. Configure Firebase credentials and verify communication between the Raspberry Pi and mobile application.
10. Test camera connectivity, face recognition, weapon detection, door lock operation, live video streaming, and notification services before deployment.
11. Ensure adequate lighting conditions around the entrance for accurate face recognition performance.
12. Protect all hardware components from excessive heat, dust, moisture, and direct environmental exposure.
13. Perform a final system inspection and functionality test before putting the system into regular operation.

Chapter 6: User Guide

a. System Startup

1. Connect the Raspberry Pi, cameras, relay module, and power supply.
2. Ensure that the Raspberry Pi and mobile device are connected to the same network.
3. Turn on the Raspberry Pi and wait for the operating system to boot.
4. Run the surveillance application using the command: python main.py
5. Verify that both cameras are connected and streaming correctly.

b. Mobile Application Login

1. Open the ENOLA Security mobile application.
2. Enter the registered email and password.
3. Press the Login button.
4. After successful authentication, the dashboard screen will be displayed.

c. Live Monitoring

1. From the dashboard, select Open Live Stream.
2. The application will display the live video feed received from the Raspberry Pi.
3. Users can switch between available camera streams using the camera selection buttons.

d. Face Recognition Process

1. An authorized person stands in front of the face recognition camera.
2. The system captures the face image and compares it with stored facial encodings.
3. If a match is found, the user is marked as authorized.
4. If no match is found, an intrusion alert is generated.

e. Weapon Detection Process

1. The second camera continuously monitors the monitored area.
2. The YOLOv8 model analyzes incoming video frames.
3. If a handgun or weapon is detected, the system generates an intrusion alert and keeps the door locked.

f. Door Access Logic

The system follows the following decision rules:

Unlock Condition

- Authorized face detected.
- No weapon detected.

Result:

- Door unlocks.
- Unlock notification is sent to Firebase and the mobile application.
- Door automatically locks again after 5 seconds.

Lock Condition

- Unknown face detected.
- Weapon detected.
- No authorized face available.

Result:

- Door remains locked.
- Security alert is generated.

g. Manual Door Control

1. Open the dashboard.
2. Press the lock/unlock button.
3. The command is sent through Firebase to the Raspberry Pi.
4. The relay updates the physical door lock state.

h. Alert Monitoring

1. Open the **Alerts** section.
2. View all security notifications including:
    - Unknown face detection
    - Weapon detection
    - Door unlocked events

- System notifications

3. Alerts are updated in real time through Firebase.

I. System Shutdown

1. Press Q on the Raspberry Pi terminal window.
2. Wait for the application to close.
3. The cameras, relay, and GPIO resources will be released automatically.
4. Shut down the Raspberry Pi safely using: sudo shutdown now

6.2 Troubleshooting Guide

Table 8: Troubleshooting Guide

Problem	Possible Cause	Solution
Camera not detected	Loose connection	Reconnect USB/IP camera
Face recognition failure	Poor lighting	Improve lighting conditions
Weapon detection inaccurate	Low image quality	Adjust camera positioning
Relay not activating	GPIO connection issue	Verify GPIO17 wiring
Lock not unlocking	12V supply failure	Check lock power supply
Firebase not syncing	Internet issue	Verify Wi-Fi connection
Mobile app not updating	Firebase delay	Restart application
Slow system performance	High CPU usage	Reduce background processes
Raspberry Pi overheating	Poor ventilation	Improve cooling system
Unauthorized access not logged	Database sync issue	Restart Firebase service

Chapter 7: Deliverables and Cost

This chapter presents the final project deliverables, project planning methodology, work distribution, project management structure, bill of materials, and the overall cost analysis of the AI-Based Smart Surveillance System. The chapter reflects the management and organizational aspects of the project along with the technical resources required for successful implementation.

The project involved both hardware and software development, requiring coordinated teamwork, proper scheduling, procurement planning, and systematic execution throughout FYP-I and FYP-II.

7.1 Deliverables

The following deliverables were completed and submitted as part of the AI-Based Smart Surveillance System project.

a. Hardware Deliverables

- Raspberry Pi 4 integrated surveillance system
- Relay-controlled electric strike lock mechanism
- Camera integration setup
- Power supply and GPIO interfacing hardware
- Fully operational prototype hardware setup

b. Software Deliverables

- Real-time face recognition module
- YOLOv8 weapon detection module
- Multi-threaded Python backend
- Firebase cloud integration
- Event logging system
- Mobile application frontend
- GPIO automation logic

c. Documentation Deliverables

- FYP-I Proposal Report
- FYP-II Final Thesis Report

- Mid Defense Presentation
- Final Defense Presentation
- Engineering Management Case Study
- System architecture diagrams
- Testing and evaluation reports
- User guide documentation

d. Functional Deliverables

The completed system successfully provides:

- Real-time face authentication
- Weapon detection
- Automated door access control
- Mobile-based monitoring
- Cloud synchronization
- Real-time notifications
- Event logging

7.2 Project Plan

The project was executed in multiple phases during FYP-I and FYP-II. A structured project management approach was followed to ensure timely completion of hardware procurement, AI model development, cloud integration, testing, and documentation.

The project planning process included:

- Requirement analysis
- Hardware procurement
- AI model training
- Embedded deployment
- Cloud integration
- Hardware-software synchronization
- Testing and validation
- Documentation and presentation preparation

The project execution timeline was divided into sequential phases to maintain smooth workflow and efficient resource utilization.

Chapter 7: Deliverables and Cost

Table 9: Project Phases

Phase	Description
Phase 1	Problem identification and literature review
Phase 2	Requirement analysis and project planning
Phase 3	Hardware procurement and Raspberry Pi setup
Phase 4	Face recognition model development
Phase 5	Weapon detection model training
Phase 6	Firebase and mobile integration
Phase 7	Hardware program development and integration
Phase 8	System testing and debugging
Phase 9	Documentation and final deployment

The project plan ensured systematic progress throughout both semesters.

7.3 Work Breakdown Structure

The project tasks were divided into manageable sub-components to improve workflow organization and team coordination.

Work Breakdown Structure (WBS)

a. Project Planning

- Problem definition
- Literature review
- Requirement gathering
- Client meetings

Chapter 7: Deliverables and Cost

b. Hardware Implementation

- Raspberry Pi setup
- Relay module integration
- Electric strike lock installation
- Camera setup
- Power supply configuration

c. AI Development

- Dataset collection
- Face recognition training
- YOLOv8 training
- Model optimization
- Embedded deployment

d. Software Development

- Python backend development
- Multi-threaded processing
- GPIO control programming
- Firebase integration
- Mobile application development

e. Testing and Validation

- Hardware testing
- Face recognition testing
- Weapon detection testing
- Cloud synchronization testing
- Performance evaluation

f. Documentation

- Report writing
- Presentation preparation
- User guide development
- Final thesis compilation

Chapter 7: Deliverables and Cost

The work breakdown structure helped organize the project into clearly manageable engineering activities.

7.4 Gantt Chart

The project activities were scheduled across both semesters using a Gantt-chart-based planning structure. The schedule ensured balanced workload distribution and proper milestone tracking.

Table 10: Project Timeline Summary

Activity	Duration
Literature Review	Weeks 01 – 03
Requirement Analysis	Weeks 04 – 05
Hardware Procurement	Weeks 06 – 07
Raspberry Pi Configuration	Weeks 07 – 08
Face Recognition Development	Weeks 08 – 10
Weapon Detection Training	Weeks 11 – 14
Firebase Integration	Weeks 15 – 16
Mobile Application Development	Weeks 15 – 22
Hardware Integration	Weeks 23 – 27
System Testing	Weeks 27 – 28
Documentation and Final Report	Weeks 28 – 32

The Gantt chart was continuously updated throughout the project lifecycle to reflect completed and pending tasks.

Chapter 7: Deliverables and Cost

The scheduling process helped:

- Track project progress
- Identify delays
- Maintain coordination
- Ensure timely deliverables

7.5 Human Resource Management

The project was completed through collaborative teamwork among all group members. Tasks were distributed according to technical expertise and project requirements to ensure efficient implementation.

Table 11: Team Member Responsibilities

Team Member	Responsibilities
Asad Mehmood (22L-6270)	Firebase integration, mobile application, System integration, YOLOv8 weapon detection
Malik Mahad Asif (22L-7670)	Face recognition implementation, Raspberry Pi integration, AI deployment, testing
Hafiz Farrukh Ahmad (22L-7733)	Hardware Program development, hardware integration, testing, documentation

Supervisor Responsibilities

Advisor:

- Technical guidance
- Project monitoring
- Progress evaluation
- Documentation review
- Engineering consultation

Chapter 7: Deliverables and Cost

Client Responsibilities

Client:

- Requirement specification
- Functional verification
- Feedback and review
- Deployment expectations

The team maintained regular meetings and collaborative coordination throughout the project to ensure smooth execution and timely completion.

7.6 Bill of Materials

Table 12: Bill of Materials

Sr. No	Component	Quantity	Purpose
1	Raspberry Pi 4	1	Main processing unit
2	MicroSD Card	3	Operating system and storage
3	USB/IP Camera	2	Video acquisition
4	Relay Module	1	Lock switching control
5	Electric Strike Lock	1	Automated door access
6	5V Power Supply	1	Raspberry Pi power
7	12V Power Supply	1	Lock power
8	Jumper Wires	Multiple	GPIO connections
9	Breadboard/Connectors	As required	Hardware interfacing
10	Wi-Fi Connectivity	1	Cloud communication

Chapter 7: Deliverables and Cost

Additional software resources included:

- Firebase cloud services
- Python libraries
- YOLOv8 framework
- Dlib library
- OpenCV

7.7 Project Cost Analysis

The overall project cost was estimated based on hardware procurement, power supplies, connectivity components, and miscellaneous implementation materials.

Table 13: Cost Breakdown

Component	Estimated Cost (PKR)
Raspberry Pi 4	35,000
MicroSD Card	2,500
Camera Module	5,000
Relay Module	800
Electric Strike Lock	4,500
5V Power Supply	2,000
12V Power Supply	2,500
Jumper Wires and Connectors	1,000
Miscellaneous Components	2,000

Estimated Total Cost

Cost (PKR)
Approximately 55,000 – 60,000 PKR

Chapter 7: Deliverables and Cost

Software Cost

Most software tools and frameworks used in the project were open-source and freely available, including:

- Python
- OpenCV
- Dlib
- YOLOv8
- Firebase basic services

This significantly reduced the overall development cost.

Cost Effectiveness

The proposed system provides:

- AI-based intelligent surveillance
- Automated access control
- Cloud monitoring
- Embedded edge processing

at a comparatively lower cost than many commercial surveillance solutions.

The low-cost implementation makes the system suitable for:

- Homes
- Offices
- Educational institutions
- Small-scale secure facilities
- Smart infrastructure applications

The project successfully demonstrates the feasibility of developing an affordable intelligent surveillance system using embedded AI and IoT technologies.

Chapter 8: Conclusion

This chapter summarizes the achievements of the AI-Based Smart Surveillance System and discusses possible future improvements and expansion opportunities. The project successfully demonstrated the integration of artificial intelligence, embedded systems, cloud communication, and automated hardware control into a unified intelligent security framework.

8.1 Conclusion

The AI-Based Smart Surveillance System was successfully designed, implemented, and tested as an intelligent real-time security and access control solution. The project addressed the limitations of traditional surveillance systems by integrating automated face recognition, weapon detection, cloud synchronization, and hardware-based access control into a single embedded platform.

The system utilized Raspberry Pi 4 as the central processing unit to execute both face recognition and weapon detection models in real time. The Dlib-based face recognition module successfully identified authorized individuals using deep learning facial embeddings, while the YOLOv8 object detection framework effectively detected suspicious objects such as guns and knives. Both AI modules operated simultaneously through a multi-threaded processing architecture, enabling efficient real-time surveillance performance.

The implemented AND-logic authentication mechanism improved security by ensuring that door access was granted only when:

- An authorized face was detected
- No weapon was identified in the environment

The project also successfully integrated:

- Relay-controlled electric strike lock automation
- GPIO-based hardware control
- Firebase cloud communication
- Real-time event logging
- Mobile application monitoring and alerts

Extensive testing confirmed stable operation of the complete system including:

- Embedded AI processing
- Hardware automation
- Real-time cloud synchronization
- Mobile application functionality
- Relay and lock response

Chapter 8: Conclusion

The system demonstrated that low-cost embedded hardware can effectively support intelligent surveillance and automated security applications without requiring expensive cloud-based infrastructure. The project successfully transformed conventional passive surveillance into an active intelligent security system capable of automated decision-making and physical access control.

Overall, the proposed AI-Based Smart Surveillance System achieved the primary objectives defined during FYP-I and extended them significantly during FYP-II through full hardware integration, cloud connectivity, mobile application support, and real-time operational deployment.

8.2 Future Improvements

Although the system achieved successful real-time operation and reliable performance, several improvements can further enhance system efficiency, scalability, accuracy, and security.

a. Facial Anti-Spoofing Implementation

Future versions of the system can include advanced anti-spoofing mechanisms capable of detecting:

- Printed photographs
- Mobile screen images
- Video replay attacks
- 3D mask attempts

This would significantly improve authentication security.

b. Improved Low-Light Performance

Detection accuracy under poor lighting conditions can be enhanced through:

- Infrared cameras
- Night-vision support
- Image enhancement algorithms
- Adaptive brightness correction

c. Advanced Threat Detection

The current system mainly focuses on weapon detection. Future improvements may include:

- Suspicious behavior analysis
- Human activity recognition
- Intrusion prediction
- Emotion recognition
- Crowd analysis

using advanced AI models.

d. Larger Face Database Optimization

Database scalability can be improved to support:

- Large-scale user management
- Faster facial embedding search
- Cloud-based user synchronization
- Enterprise-level deployments

e. Improved Embedded Performance

The system performance can be enhanced using:

- Hardware AI accelerators
- Tensor Processing Units (TPUs)
- NVIDIA Jetson platforms
- Optimized lightweight AI models

This would improve inference speed and support more advanced AI tasks.

f. Multi-Camera Integration

Future systems may support:

- Multiple surveillance cameras
- Centralized monitoring
- Multi-angle detection
- Wider surveillance coverage

for large-scale environments.

g. Enhanced Mobile Application

The mobile application can be upgraded by adding:

- Voice commands
- Real-time analytics dashboards
- Video recording playback
- Multi-user management
- Advanced notification customization

8.3 Future Scope

The future scope of the AI-Based Smart Surveillance System is broad due to the increasing global demand for intelligent automation, AI-powered security infrastructure, and smart monitoring systems.

The proposed framework can be expanded for deployment in:

- Smart homes
- Offices
- Banks
- Laboratories
- Airports
- Educational institutions
- Government buildings
- Industrial facilities

Chapter 8: Conclusion

Future versions of the system can become part of larger smart city infrastructures by integrating:

- IoT sensor networks
- Centralized surveillance systems
- Smart emergency response mechanisms
- Automated law enforcement monitoring

The system also provides a strong foundation for research and development in:

- Embedded AI systems
- Edge computing
- Intelligent surveillance
- Real-time threat analysis
- Automated access control

Integration with cloud-based AI services and advanced machine learning algorithms may further improve:

- Detection accuracy
- Predictive security analysis
- Behavioral monitoring
- Autonomous decision-making

Additionally, future commercial implementations may include:

- Subscription-based cloud services
- Enterprise security management
- Smart building automation
- AI-assisted public safety systems

The project demonstrates the practical feasibility of developing scalable and affordable intelligent surveillance solutions using embedded AI technologies. As artificial intelligence and IoT technologies continue to evolve, systems like the proposed AI-Based Smart Surveillance System are expected to play a major role in the development of future smart security infrastructures.

Appendices

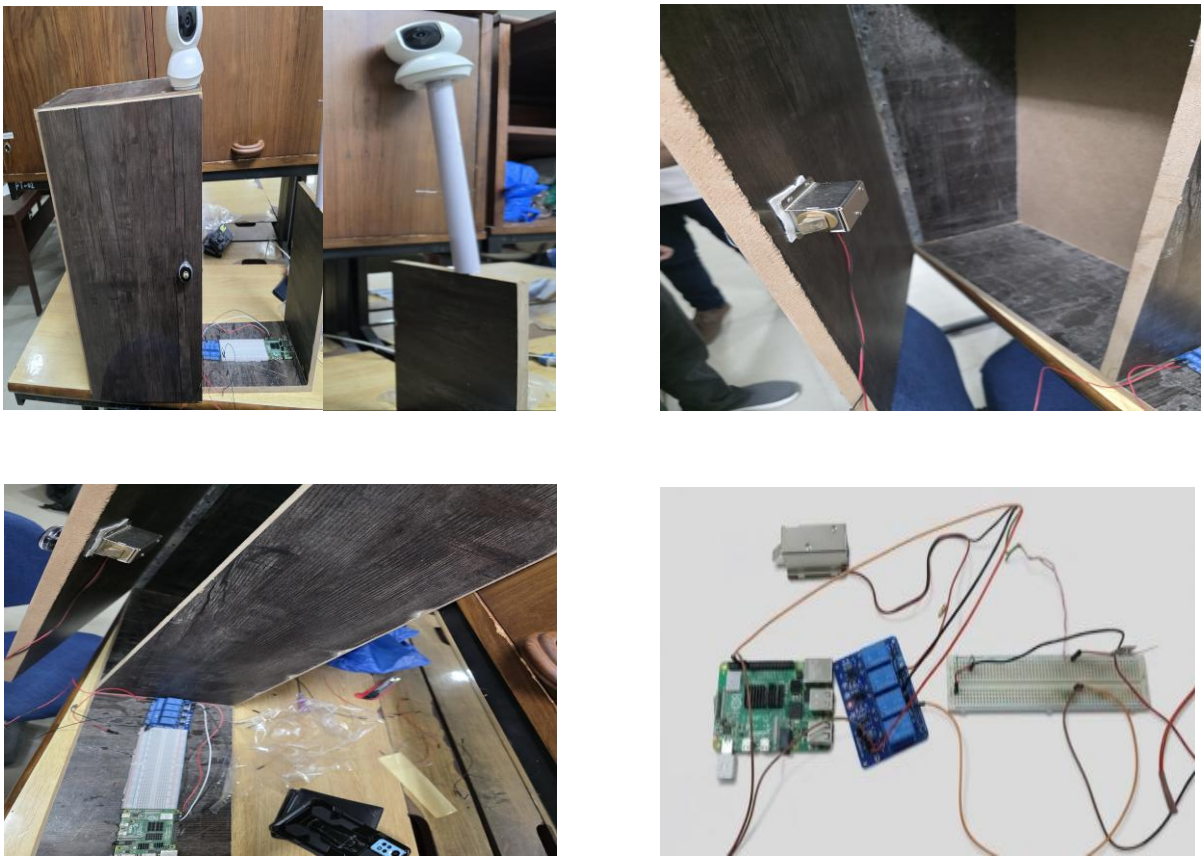


Figure 12: System Prototype

Appendices

```
import cv2
import dlib
import pickle
import numpy as np
from pathlib import Path
from ultralytics import YOLO
import RPi.GPIO as GPIO
import firebase_admin
from firebase_admin import credentials, db
import time
import threading
from collections import deque


# CONFIGURATION
RELAY_PIN=17
UNLOCK_SECONDS=5
FRAME_SKIP=6
FACE_SCALE=0.35
YOLO_IMGSZ=96
YOLO_CONF=0.5
FACE_THRESHOLD=0.5
ALERT_COOLDOWN=15
DECISION_FRAMES=5
RTSP_BUFFER=1
```

```
BASE_DIR=Path(__file__).resolve().parent
MODELS_DIR=BASE_DIR/"models"


RTSP_FACE="rtsp://asad016.am:12345678@10.139.49.7:554/stream1"
RTSP_OBJECT="rtsp://asad012.am:12345678@10.139.49.65:554/stream2"


# FIREBASE
cred=credentials.Certificate('/home/asad99/enola-firebase-key.json')


firebase_admin.initialize_app(cred,{
    'databaseURL':'https://enola-24901-default-rtdb.firebaseio.com'
})


# GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)


# ACTIVE LOW RELAY
GPIO.setup(RELAY_PIN,GPIO.OUT,initial=GPIO.HIGH)
```

```
# LOAD MODELS
print("[INIT] Loading models...")


detector=dlib.get_frontal_face_detector()


shape_pred=dlib.shape_predictor(
    str(MODELS_DIR/"shape_predictor_5_face_landmarks.dat")
)


face_rec_model=dlib.face_recognition_model_v1(
    str(MODELS_DIR/"dlib_face_recognition_resnet_model_v1.dat")
)


yolo=YOLO(str(MODELS_DIR/"best.pt"))
yolo.to('cpu')


# FACE ENCODINGS
with open(MODELS_DIR/"face_encodings_dlib.pkl","rb") as f:
```

```
    known_encodings,known_names=pickle.load(f)

known_encodings=np.array(
    known_encodings,
    dtype=np.float32
)

WEAPON_LABELS={
    "gun",
    "handgun",
    "pistol",
    "rifle",
    "weapon",
    "knife"
}

print(f"[INIT] Loaded {len(known_names)} known faces")

# CAMERA
def open_camera(url):

    cap=cv2.VideoCapture(url,cv2.CAP_FFMPEG)

    cap.set(cv2.CAP_PROP_BUFFERSIZE,RTSP_BUFFER)
    cap.set(cv2.CAP_PROP_FRAME_WIDTH,320)
    cap.set(cv2.CAP_PROP_FRAME_HEIGHT,240)
    cap.set(cv2.CAP_PROP_FPS, 15)

    return cap

cap_face=open_camera(RTSP_FACE)
cap_object=open_camera(RTSP_OBJECT)

if not cap_face.isOpened():
    print("❌ Face camera failed")
    GPIO.cleanup()
    exit()

if not cap_object.isOpened():
    print("❌ Object camera failed")
    GPIO.cleanup()
    exit()

print("✅ ENOLA AI Security System Started")

# STATE
last_decision=None
relay_lock=threading.Lock()
relock_timer=None
_last_alert_times={}
decision_history=deque(maxlen=DECISION_FRAMES)
_alert_queue=deque()
_alert_queue_lock=threading.Lock()

# FIREBASE WORKER
def firebase_worker():

    while True:

        try:

            with _alert_queue_lock:

                if _alert_queue:
                    task=_alert_queue.popleft()
                else:
                    task=None

            if task:
                task()
            else:
                time.sleep(0.1)

        except Exception as e:
            print(f"[Firebase Worker] {e}")

threading.Thread(
    target=firebase_worker,
    daemon=True
).start()
```

```
# FIREBASE HELPERS
def push_alert(title,message,alert_type):
    def write():
        try:
            db.reference('alerts').push({
                'title':title,
                'message':message,
                'type':alert_type,
                'timestamp':int(time.time()*1000),
                'read':False
            })
            print(f"[ALERT] {title}")
        except Exception as e:
            print(f"[Firebase ERROR] {e}")
    with _alert_queue_lock:
        _alert_queue.append(write)

def update_door_status(status):
    def write():
        try:
            db.reference('door/status').set(status)
        except Exception as e:
            print(f"[Firebase ERROR] {e}")
    with _alert_queue_lock:
        _alert_queue.append(write)

def cooldown_push(title,message,alert_type):
    now=time.time()
    if now-_last_alert_times.get(alert_type,0)>=ALERT_COOLDOWN:
        _last_alert_times[alert_type]=now
        push_alert(title,message,alert_type)

# GPIO FUNCTIONS
def do_relock():
    with relay_lock:
        GPIO.output(RELAY_PIN, GPIO.HIGH)
    update_door_status('locked')
    print("[GPIO] Auto Relocked")

def trigger_unlock(name):
    global relock_timer
    with relay_lock:
        GPIO.output(RELAY_PIN, GPIO.LOW)
    update_door_status('unlocked')
    push_alert(
        '✅ Door Unlocked',
        f'Access granted: {name}',
        'system'
    )
    print(f"[GPIO] UNLOCKED - {name}")
    if relock_timer and relock_timer.is_alive():
        relock_timer.cancel()
    # AUTO LOCK AFTER 5 SECONDS
```

```
    relock_timer=threading.Timer
(
        UNLOCK_SECONDS,
        do_relock
    )

    relock_timer.daemon=True
    relock_timer.start()

def trigger_lock():

    with relay_lock:
        GPIO.output(RELAY_PIN,
GPIO.HIGH)

    update_door_status('locked')

    print("[GPIO] LOCKED")

# FLUTTER COMMAND
LISTENER
def listen_app_commands():

    def on_command(event):

        data=event.data

        if not isinstance(data,dict):
            return

        command=data.get('comma
nd','')

        print(f"[APP CMD]
{command}")

        if command=='unlock':
            trigger_unlock('Mobile
App')

        elif command=='lock':
            trigger_lock()

    db.reference('door/control').lis
ten(on_command)

threading.Thread(
    target=listen_app_commands,
    daemon=True
).start()

print("✅ Firebase Listener
Started")

# FACE DETECTION
def detect_face(frame_bgr):

    small=cv2.resize(
        frame_bgr,
        (0,0),
        fx=FACE_SCALE,
        fy=FACE_SCALE
    )

    rgb=cv2.cvtColor(
        small,
        cv2.COLOR_BGR2RGB
    )

    faces=detector(rgb,0)

    face_verified=False
    verified_name="UNKNOWN
"

    for face in faces:

        landmarks=shape_pred(rgb,
face)

        encoding=np.array(
            face_rec_model.compute
_face_descriptor(
                rgb,
                landmarks
            ),
            dtype=np.float32
        )

        distances=np.linalg.norm(
            known_encodings-
encoding,
            axis=1
        )
```

```
        min_dist=float(np.min(dist
ances))
        best_idx=int(np.argmin(dist
ances))

        if
min_dist<FACE_THRESHOLD
:

            face_verified=True
            verified_name=known_n
ames[best_idx]

        else:

            cooldown_push(
                '⚠️ Unknown Face',
                'Unknown person
detected',
                'intrusion'
            )

    return
face_verified,verified_name

# OBJECT DETECTION
def detect_objects(frame_bgr):

    gun_detected=False

    results=yolo(
        frame_bgr,
        imgsz=YOLO_IMGSZ,
        conf=YOLO_CONF,
        stream=True,
        verbose=False
    )

    for r in results:

        for box in r.boxes:

            label=yolo.names[
                int(box.cls[0])
            ].lower()

            if label in
WEAPON_LABELS:

                gun_detected=True

                cooldown_push(
                    '🔴 Weapon
Detected',
                    f'{label} detected at
entry',
                    'intrusion'
                )

    return gun_detected

# CAMERA BUFFER
CLEANER
def grab_latest(cap):
    ret,frame=cap.read()

    if not ret:
        return False,None

    for _ in
range(RTSP_BUFFER):
        cap.grab()

    return True,frame

# MAIN LOOP
frame_count=0

try:

    while True:

        ret1,frame_face=grab_lates
t(cap_face)
        ret2,frame_object=grab_lat
est(cap_object)

        if not ret1 or not ret2:
            time.sleep(0.05)
            continue

        frame_count+=1

        # RUN AI EVERY N
FRAMES
```

```
        if frame_count%FRAME_SKIP==0:

            face_verified,verified_name=detect_face(frame_face)

            gun_detected=detect_objects(frame_object)

            # FINAL DECISION
            raw_decision="UNLOCK" if (
                face_verified and not gun_detected
            ) else "LOCK"

            decision_history.append(raw_decision)

            # DEBOUNCE
            if len(decision_history)==DECISION_FRAMES and \
               len(set(decision_history))==1:

                decision=decision_history[0]

                if decision!=last_decision:

                    last_decision=decision

                    if decision=="UNLOCK":
                        trigger_unlock(verified_name)
                    else:
                        trigger_lock()

                print(f"[STATUS] {last_decision}")

            # SMALL DISPLAY FOR VNC
            small_face=cv2.resize(frame_face,(320,240))
            small_object=cv2.resize(frame_object,(320,240))

            cv2.imshow("Face Camera",small_face)
            cv2.imshow("Object Camera",small_object)

            # PRESS Q TO EXIT
            if cv2.waitKey(1)&0xFF==ord('q'):
                print("🛑 Exiting System")
                break

    except KeyboardInterrupt:
        print("🛑 Keyboard Interrupt Detected")

    # CLEANUP
    finally:
        cap_face.release()
        cap_object.release()
        cv2.destroyAllWindows()

        if relock_timer and relock_timer.is_alive():
            relock_timer.cancel()

        GPIO.cleanup()

        print("✅ System shutdown cleanly")
```

Glossary

List all acronyms and technical terms in alphabetical order along with their brief description, as shown below:

map	Mean Average Precision
CNN	Convolutional neural network
YOLO	You Only Looks Once
Dlib	Digital Library (Machine Learning Toolkit)
RTSP	Real-Time Streaming Protocol
FaceNet	Face Embedding Network for Face Recognition
SDK	Software Development Kit

Final Year Project Contribution Form

Contributor Role Taxonomy

No.	Contribution	Description
1.	Conceptualization	Developing initial project ideas, goals, plans, and schedules.
2.	Problem Analysis	Conducting background research, literature review, and analysis to assess the project's feasibility.
3.	Methodology	Designing methodology and developing prototypes / models.
4.	Project Administration	Managing the planned project activity and its execution.
5.	Resource Acquisition	Gathering necessary materials, hardware modules, and computing resources.
6.	Hardware Implementation	Participating in the hardware implementation phase.
7.	Software Implementation	Developing computer programs/software to implement the algorithms.
8.	Validation	Conducting experiments and tests to verify the project's intended output.
9.	Documentation	Writing project report and preparing other documents.
10.	Presentation	Preparing and participating in project presentations to communicate the progress.

Each group member is required to list his/her contribution(s) in the completion of the FYP using the taxonomy given above.

Group Member	Group Member's Contribution	Signatures
Asad Mehmood	1,2,3,4,5,6,7,10,9,8	
Malik Mahad Asif	2,5,10,9,3,8	
Hafiz Farukh Ahmad	2,6,8,9,10,5	

Supervisor's Signature _____

Supervisor's Name _____

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	22L-7733	Hafiz Farrukh Ahmed	22	EE	LHR
Supervisor(s)	Umer Altaf				
Summary: (Min 50 - Max 100 words)	This project presents an AI-Based Smart Surveillance System that combines real-time face recognition, weapon detection, automated access control, and cloud monitoring. The system is implemented on a Raspberry Pi 4 using Dlib CNN-based face recognition and YOLOv8 object detection. Access is granted only when an authorized face is recognized and no weapon is detected. Firebase integration enables real-time notifications, event logging, and remote monitoring through a Flutter mobile application. The system achieved 97% face recognition accuracy and 93% weapon detection accuracy while providing a low-cost and scalable smart security solution.				
Introduction (One line)	An AI-powered surveillance and access control system that enhances security through real-time face recognition, weapon detection, and automated door control.				
Problem Statement (Max 2 lines)	Traditional CCTV and access control systems rely heavily on human monitoring and are vulnerable to unauthorized access. An intelligent, automated, and low-cost security solution capable of real-time threat detection and access control is required.				
Objective	To design, develop, and validate an AI-powered surveillance system capable of real-time face recognition, weapon detection, automated door access control, and cloud-based remote monitoring.				
Result(s) & Conclusions	The system achieved **97% face recognition accuracy** and **93% weapon detection accuracy** with reliable real-time operation on Raspberry Pi 4. Integrated testing confirmed successful automated access control, Firebase synchronization, mobile notifications, and stable hardware performance, demonstrating the feasibility of an intelligent low-cost surveillance solution.				
Future Directions	➢ ➤ Facial anti-spoofing for enhanced security ➤ Multi-camera surveillance support ➤ Advanced behavior and threat analysis ➤ Solar-powered operation for energy efficiency ➤ Cloud-based analytics and reporting dashboard ➤ Large-scale deployment with centralized management				



